

诚信应考，考试作弊将带来严重后果！

华南理工大学本科生期末考试

2024-2025-2 学期章节试卷

《Calculus Chapter Review-Double integral and Triple integral》

- 注意事项：1. 开考前请将密封线内各项信息填写清楚；
2. 所有答案请直接答在试卷上；
3. 考试形式：闭卷；
4. 本试卷共 5 大题，满分 50 分，考试时间 50 分钟。

1, Evaluate $\int_0^1 \int_0^{\sqrt{1-y^2}} \sin(x^2 + y^2) dy dx$

2, Changing the order of integration $\int_{\frac{1}{2}}^1 \int_{x^3}^x f(x, y) dy dx$

3, Changing the order of integration $\int_0^2 \int_{y^2}^{2y} f(x, y) dx dy$

4, Evaluate iterated integrals: $\int_{\frac{1}{4}}^{\frac{1}{2}} dy \int_{\frac{1}{2}}^{\sqrt{y}} e^{\frac{y}{x}} dx + \int_{\frac{1}{2}}^1 dy \int_y^{\sqrt{y}} e^{\frac{y}{x}} dx$

5, Find $\iint_D \frac{y}{x+y} e^{(x+y)^2} dx dy$, D is bounded by $x+y=1$, $x=0$ and $y=0$.

6, Calculate $\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_{-\sqrt{9-x^2-z^2}}^{\sqrt{9-x^2-z^2}} (x^2 + y^2 + z^2)^{3/2} dy dz dx$

7, Calculate $\iiint_{\Omega} (x^2 + y^2) dV$, Ω is bounded by the planes $z=2$, $z=8$ and surface that results when the parabola $y^2 = 2z$ in the yz -plane is revolved about the z -axis.

8, Calculate $\int_0^3 \int_0^{\sqrt{9-x^2}} \int_0^2 (x^2 + y^2)^{1/2} dz dy dx$